

SIDE PROJECTS

DIBENDU PAL

SCHUR' S THEOREM

In 1905, I.Schur proved that the maximum number of linearly independent mutually commuting complex matrices of order n is $\lfloor n^2/4 \rfloor + 1$. We present a simpler proof here given by M.Mirzakhani using induction on n . Within the proof, we will use a theorem from the Linear Algebra book of Hoffmen and Kunze, which we mention below.

Theorem 1. *Let $\mathcal{F}$ be a commuting family of $n \times n$ matrices over an algebraically closed field F . Then there exists a non-singular $n \times n$ matrix P with entries in F such that $P^{-1}AP$ is upper-triangular, for every matrix A in $\mathcal{F}$.*

Theorem 2. *The maximum number of mutually commuting linearly independent matrices of order n over $\mathbb{C}$ is $\lfloor n^2/4 \rfloor + 1$.*

Proof. We will prove using induction on n , that the maximum number of linearly independent matrices of order n over $\mathbb{C}$ is at most $\lfloor n^2/4 \rfloor + 1$. For $n = 1$, it is trivial. Assume that the theorem is true for $n = 1$. Let $\mathcal{F}$ be a commuting family of $n \times n$ matrices over F with more than $\lfloor n^2/4 \rfloor + 1$ linearly independent matrices. Since, $\mathbb{C}$ is algebraically closed, there exists a nonsingular matrix P with entries in F such that $P^{-1}\mathcal{F}P$ is an upper-triangular family of matrices by previous theorem. Let us denote the vectorspace spanned by the family $P^{-1}\mathcal{F}P$ by V . Since, $\mathcal{F}$ is commuting, V consists of commuting upper-triangular matrices. Now, assume that $\dim V \geq \lfloor n^2/4 \rfloor + 2$. Let

$$\{M_1, M_2, \dots, M_{\lfloor n^2/4 \rfloor + 2}\}$$

be any linearly independent subset of V . Since for each $i = 1, \dots, \lfloor n^2/4 \rfloor + 2$, M_i is upper-triangular, there exists a $(n-1) \times (n-1)$ matrix A_i and a $1 \times (n-1)$ matrix N_i and single element n_i such that

$$M_i = \left[\begin{array}{c|c} n_i & (N_i)_{1 \times (n-1)} \\ \hline 0 & (A_i)_{(n-1) \times (n-1)} \end{array} \right].$$

We also observe that $A_i A_j = A_j A_i$, since M_i 's commute in V for all $1 \leq i, j \leq \lfloor n^2/4 \rfloor + 2$. Define W as the vector space by

$$W = \text{Span} \{A_1, A_2, \dots, A_{\lfloor n^2/4 \rfloor + 2}\}.$$

Let $k = \dim W$. Using the induction hypothesis, we have $k \leq \lfloor (n-1)^2/4 \rfloor + 1$. Without loss of generality, assume that W is spanned by the linearly independent matrices $A_1, A_2, \dots, A_k$. Hence for each i , we get $a_{i1}, a_{i2}, \dots, a_{ik} \in \mathbb{C}$ such that

$$A_i = \sum_{j=1}^k a_{ij} A_j.$$

Now define,

$$C_i = M_i - \sum_{j=1}^k a_{ij} M_j \quad \text{for } i > k.$$

Clearly we notice that C_i 's are linearly independent since M_i 's are linear independent for each i . Also, for any $i = k+1, \dots, \lfloor n^2/4 \rfloor + 2$,

$$C_i = \begin{bmatrix} \dots * \dots \\ \hline (0)_{(n-1) \times n} \end{bmatrix} = \begin{pmatrix} v_i \\ 0 \end{pmatrix},$$

where v_i is a $1 \times n$ matrix in V . Furthermore the vectors v_i are linearly independent since C_i 's are so.

Now, we exchange the matrices position as given after. Since for each i , we have M_i upper-triangular, there exists a $(n-1) \times (n-1)$ matrix A'_i , and a $(n-1) \times 1$ matrix N'_i and single element n'_i such that

$$M'_i = \left[\begin{array}{c|c} (A'_i)_{(n-1) \times (n-1)} & N'_i \\ \hline 0 & n'_i \end{array} \right].$$

By a similar reasoning we obtain a set of linearly independent $n \times 1$ matrices, $v'_{s+1}, \dots, v'_{\lfloor n^2/4 \rfloor + 2}$, such that $s \leq \lfloor (n-1)^2/4 \rfloor + 1$, and for each i ,

$$C'_i = \left[\begin{array}{c|c} (0)_{n \times (n-1)} & \begin{matrix} \vdots \\ * \\ \vdots \end{matrix} \end{array} \right] = \begin{pmatrix} 0 & v'_i \end{pmatrix}$$

is a matrix in V . Since all C_i 's and C'_i 's belong to the same commuting family V , we deduce that

$$C_i C'_j = C'_j C_i = 0,$$

which implies from left side,

$$v_i v'_j = 0$$

for $i = k + 1, \dots, \lfloor n^2/4 \rfloor + 2$ and $j = s + 1, \dots, \lfloor n^2/4 \rfloor + 2$.

Let X be the $(\lfloor n^2/4 \rfloor - k + 2) \times n$ matrix with its i -th row as v_i . Since v_i 's are linearly independent, we have

$$\text{rank} X = \lfloor n^2/4 \rfloor - k + 2.$$

On the other hand, $X v'_j = 0$ for $j = s + 1, \dots, \lfloor n^2/4 \rfloor + 2$, where v'_j 's are linearly independent. Hence we get,

$$\text{nullity} X \geq \lfloor n^2/4 \rfloor - s + 2.$$

Since $\text{rank} X + \text{nullity} X = n$, we get,

$$\begin{aligned} n &\geq 2 \left\lfloor \frac{n^2}{4} \right\rfloor + (-k + 1) + (-s + 1) + 2 \\ &\geq 2 \left(\left\lfloor \frac{n^2}{4} \right\rfloor - \left\lfloor \frac{(n-1)^2}{4} \right\rfloor + 1 \right) \\ &\geq 2 \left\lfloor \frac{n}{2} \right\rfloor + 2 \end{aligned}$$

where last line comes if we consider both even and odd cases. Hence we get a contradiction.

To prove it is maximum we consider the following example. Take the set

$$\mathcal{F} = \{E_{ij} \mid 1 \leq i \leq \lfloor n/2 \rfloor, \lfloor n/2 \rfloor + 1 \leq j \leq n\} \cup \{I\},$$

where E_{ij} 's are the elementary matrices (that forms the basis of matrices) and I is the identity matrix, is a commuting family of $n \times n$ matrices with exactly $\lfloor n^2/4 \rfloor + 1$ linearly independent matrices. Hence we proved the above theorem. □

TO PROVE THE IRREDUCIBILITY OF $x^n - x - 1$ IN $\mathbb{Q}[x]$

Define reciprocal of $f(x) = \sum_{i=0}^n a_i x^i \in \mathbb{F}[x]$ as

$$\tilde{f}(x) = a_0 x^n + a_1 x^{n-1} + \cdots + a_{n+1} x + a_n.$$

Equivalently, if degree of f is n , then we have

$$\tilde{f}(x) = x^n f(x^{-1}).$$

Some of its properties are:

- (1) If $f(0) \neq 0$ then $\deg f = \deg \tilde{f}$ and $\tilde{\tilde{f}} = f$
- (2) If $f = gh$ then $\tilde{f} = \tilde{g}\tilde{h}$
- (3) If $c \in \mathbb{F} - \{0\}$ then $c\tilde{f} = \tilde{cf}$

Lemma 3. *Let $f(x) = x^n - x - 1 \in \mathbb{Z}[x]$ for $n \geq 2$. Then $f(x)$ and $\tilde{f}(x)$ have no common root in $\mathbb{C}$.*

Proof. Note $\tilde{f}(x) = -x^n - x^{n-1} + 1$. If possible, let $\alpha \in \mathbb{C}$ be a common root. Then we have the equations

$$\alpha^n = \alpha + 1, \quad \alpha^n = -\alpha^{n-1} + 1.$$

From this we get

$$-\alpha^{n-1} = \alpha,$$

that is,

$$\alpha^n = -\alpha^2.$$

Putting this into the equation $\alpha^n = \alpha + 1$, we get $\alpha^2 + \alpha + 1 = 0$. Thus,

$$\alpha^3 = 1.$$

If we write $n = 3k + j$, where $j \in \{0, 1, 2\}$, then we see that $\alpha^n = \alpha^j$. We also get $\alpha^{n-1} = \alpha^{j-1}$. Again, substituting this into the equation $\alpha^n = \alpha + 1$, we get that

$$\alpha^j = \alpha + 1.$$

If $j = 0$ then we get $\alpha = 0$. This is not possible. If $j = 1$ then also we get a contradiction. If $j = 2$, then we get $\alpha^2 = \alpha + 1$. Also, substituting into the equation $\alpha^n = -\alpha^{n-1} + 1$, we get $\alpha^2 = -\alpha + 1$. Combining these we get $\alpha = 0$, a contradiction. This completes the proof of the Lemma. $\square$

Lemma 4. *If f is reducible in $\mathbb{Z}[x]$, then there is $k(x) \in \mathbb{Z}[x]$ with $\deg(k(x)) = \deg(f(x))$ such that $f\tilde{f} = k\tilde{k}$ and $k \notin \{\pm f, \pm \tilde{f}\}$.*

Proof. Let $f(x) = g(x)h(x)$ where $\deg(g(x)), \deg(h(x)) \geq 1$. Since $f(0) \neq 0$ both $g(0), h(0) \neq 0$. Thus, $\deg(g(x)) = \deg(\tilde{g}(x))$ and $\deg(h(x)) = \deg(\tilde{h}(x))$. Let $k(x) = g(x)\tilde{h}(x)$. Then

$$\deg(k(x)) = \deg(g(x)) + \deg(\tilde{h}(x)) = \deg(g(x)) + \deg(h(x)) = \deg(f(x)).$$

Further,

$$k\tilde{k} = (g\tilde{h})(\tilde{g}h) = (gh)(\tilde{g}\tilde{h}) = f\tilde{f}.$$

Next we show that $k \notin \{\pm f, \pm \tilde{f}\}$. We work out only the case $k = f$ as other cases are similar. If $k = f$ then

$$g\tilde{h} = gh$$

which shows that $\tilde{h} = h$. This shows that h and $\tilde{h}$ have a common root, which in turn shows that f and $\tilde{f}$ have a common root. This contradicts Lemma 3. This completes the proof of the Lemma. $\square$

Theorem 5. *Given $f(x) = x^n - x - 1 \quad \forall n \geq 2$ is irreducible in $\mathbb{Q}[x]$.*

Proof. By Gauss Lemma, it is enough to check that f is irreducible in $\mathbb{Z}[x]$. Assume f factors in $\mathbb{Z}[x]$. By Lemma 3 and Lemma 4, there is $k(x) = \sum_{i=0}^n b_i x^i \in \mathbb{Z}[x]$ such that

$$k(x) \notin \{\pm f, \pm \tilde{f}\} \text{ and } f\tilde{f} = k\tilde{k}.$$

Now

$$\begin{aligned} k(0)\tilde{k}(0) &= b_0 b_n = f(0)\tilde{f}(0) = -1 \\ &\Rightarrow b_n = \pm 1 \text{ and } b_0 = -b_n \end{aligned}$$

Replacing k by $-k$, if necessary, we may assume that $b_n = 1$. Then $b_0 = -1$. Comparing coefficients of x^n in $f\tilde{f}$ and $k\tilde{k}$ we get

$$1^2 + (-1)^2 + (-1)^2 = b_0^2 + b_1^2 + b_2^2 + \dots + b_n^2$$

This shows that $b_i = \pm 1$ for exactly one $i \in \{1, 2, \dots, n-1\}$. So $k(x) = x^n + b_i x^i - 1$. Then

$$\begin{aligned} k\tilde{k} &= (x^n + b_i x^i - 1)(-x^n + b_i x^{n-i} + 1) \\ &= -x^{2n} - b_i x^{n+i} + 3x^n + b_i x^{2n-i} - b_i x^{n-i} - 1. \end{aligned}$$

On the other hand,

$$f\tilde{f} = -x^{2n} - x^{2n-1} + x^{n+1} + 3x^n + x^{n-1} - x - 1.$$

It is clear that $2n - i > n - i$ and $n + i > n - i$. Let us assume that $n > 2$. Then $n - 1 > 1$. It follows that $n - i = 1$ and that $b_i = 1$. Thus, $i = n - 1$ and $b_{n-1} = 1$. Thus, $k(x) = x^n + x^{n-1} - 1 = -\tilde{f}(x)$. This is a contradiction. Thus, it follows that if $n > 2$ then $f(x)$ is irreducible in $\mathbb{Q}[x]$. When $n = 2$, we can easily check that the roots are not rational. This completes the proof of the Theorem. $\square$

TO PROVE e AND π ARE TRANSCENDENTAL NUMBERS

The two well known and most important irrational real numbers are e and π . Their existence were known for over 2000 years ago. But their irrationality was first proved by Lambert in 1750s. Proving that a particular number, such as e and π , is transcendental was also challenge back then. The transcendence of e was not proved until 1873, when Hermite gave a proof. Nine years later, Lindemann used Hermite's method to prove that π is transcendental. We give here a proof of an alternative version of Lindemann-Weierstrass theorem given by A. Baker which implies the transcendence of both e and π

Theorem 6. (Lindemann-Weierstrauss) *Let $\alpha_1, \dots, \alpha_m$ be distinct algebraic numbers. Then the exponentials $e^{\alpha_1}, \dots, e^{\alpha_m}$ are linearly independent over $\mathbb{Q}$.*

Proof. Step I. Suppose that there are non-zero $a_j \in \mathbb{Q}$ for $1 \leq j \leq m$, such that

$$\sum_{j=1}^m a_j e^{\alpha_j} = 0.$$

We may also assume that $a_j \in \mathbb{Z}$ for all j . Let K be the normal closure of $\mathbb{Q}(\alpha_1, \dots, \alpha_m)/\mathbb{Q}$. Then K is a Galois extension of $\mathbb{Q}$. Let, $\text{Gal}(K/\mathbb{Q}) = \{\sigma_1, \dots, \sigma_n\}$ where $\sigma_1 = \text{Id}$. Since $\sum_{j=1}^m a_j e^{\alpha_j} = 0$, we have

$$\prod_{k=1}^n \left(\sum_{j=1}^m a_j e^{\sigma_k(\alpha_j)} \right) = 0.$$

Now assume,

$$\prod_{k=1}^n \left(\sum_{j=1}^m a_j e^{\sigma_k(\alpha_j)} \right) = \sum_{j=0}^r c_j e^{\beta_j},$$

where the $c_j \in \mathbb{Z}$ and the β_j can be chosen to be distinct elements of K , where each β_j is a sum of terms of the form $\sigma_k(\alpha_l)$. Also there exists some $c_j \neq 0$; without loss of generality, say $c_0 \neq 0$. If $\sigma \in \text{Gal}(K/\mathbb{Q})$, then the n terms $\sum_{j=1}^m a_j e^{\sigma \sigma_k(\alpha_j)}$ for $1 \leq k \leq n$ are the terms $\sum_{j=1}^m a_j e^{\sigma_k(\alpha_j)}$ in some order. Hence, the product is unchanged when replacing $\sigma_k(\alpha_j)$ by $\sigma \sigma_k(\alpha_j)$. Since each β_j is a sum of terms of the form $\sigma_k(\alpha_l)$, the exponents in the expansion of $\prod_{k=1}^n \left(\sum_{j=1}^m a_j e^{\sigma \sigma_k(\alpha_j)} \right)$ are the various $\sigma(\beta_j)$. Thus, we get equations

$$c_0 e^{\sigma_0(\beta_0)} + \sum_{j=1}^r c_j e^{\sigma_j(\beta_j)} = 0$$

for each i . Multiply the i th equation by $e^{-\sigma_i(\beta_0)}$ to get

$$c_0 + \sum_{j=1}^r c_j e^{\sigma_i(\gamma_j)} = 0. \quad (1)$$

where $\gamma_j = \beta_j - \beta_0$. Note that $\gamma_j \neq 0$ since the β_j are all distinct. Since $\gamma_j \in K$, γ_j is algebraic over $\mathbb{Q}$. Thus, for a fixed j , the elements $\sigma_i(\gamma_j)$ are roots of an irreducible polynomial $g_j(x) \in \mathbb{Z}[x]$ of atmost $\deg(n)$, where the leading coefficient b_j of $g_j(x)$ can be taken to be a positive integer. This $g_j(x) = b_j \min(\mathbb{Q}, \gamma_j)$ (where b_j is the least integer multiplied to clear out the denominators from coefficients). Observe that $g_j(0) \neq 0$.

StepII. If $f(x)$ is a polynomial, let

$$F(x) = \sum_{i=0}^{\infty} f^{(i)}(x),$$

where $f^{(i)}(x)$ is the i th derivative of f . Then the sum is finite, since f is a polynomial, so F is also a polynomial. Note that $F(x) - F'(x) = f(x)$, so

$$\begin{aligned} \frac{d}{dx} (e^{-x} F(x)) &= -e^{-x} f(x) \\ \implies \int_0^a e^{-x} f(x) dx &= F(0) - e^{-a} F(a) \\ \implies F(a) - e^a F(0) &= -e^a \int_0^a e^{-x} f(x) dx. \end{aligned}$$

Put $a = \sigma_i(\gamma_j)$, and multiply by c_j , and summing over $i(1 \leq i \leq n)$ and $j(1 \leq j \leq r)$, we get

$$\begin{aligned} \sum_{j=1}^r \sum_{i=1}^n c_j F(\sigma_i(\gamma_j)) - F(0) \sum_{j=1}^r \sum_{i=1}^n c_j e^{\sigma_i(\gamma_j)} \\ = - \sum_{j=1}^r \sum_{i=1}^n c_j e^{\sigma_i(\gamma_j)} \int_0^{\sigma_i(\gamma_j)} e^{-z} f(z) dz. \end{aligned}$$

Using Equation (1) and rearranging the second sum we get

$$\begin{aligned} (7) \quad nc_0 F(0) + \sum_{j=1}^r c_j \sum_{i=1}^n F(\sigma_i(\gamma_j)) \\ = - \sum_{j=1}^r \sum_{i=1}^n c_j e^{\sigma_i(\gamma_j)} \int_0^{\sigma_i(\gamma_j)} e^{-z} f(z) dz. \end{aligned}$$

In next step we will choose certain f such that L.H.S of (2) is not divisible by p , hence absolute value greater than 1 while the R.H.S of (2) tends 0 as we take p larger. This will give our contradiction.

Step III. Now we define f by

$$f(x) = \frac{(b_1 \cdots b_r)^{prn}}{(p-1)!} x^{p-1} \left(\prod_{j=1}^r g_j(x) \right)^p,$$

where we choose $p > \max_j \{b_j, g_j(0)\}$. Observe that,

$$f(0) = f'(0) = \cdots = f^{(p-2)}(0) = 0$$

and

$$f^{(p-1)}(0) = (b_1 \cdots b_r)^{pnr} \prod_{j=1}^r g_j(0)^p \neq 0.$$

So p does not divide $f^{(p-1)}(0)$. Moreover, for $t \geq p$, the polynomial $f^{(t)}(x)$ can be written in the form

$$f^{(t)}(x) = p (b_1 \cdots b_r)^{pnr} h_t(x),$$

where $h_t(x) \in \mathbb{Q}[x]$ has degree at most $prn - 1$ and $f^{(t)}(x) \in \mathbb{Z}[x]$. Thus, $f^{(t)}(0)$ is divisible by p for $t \geq p$. Hence, $F(0) = f^{(p-1)}(0) + \sum_{j \geq p} f^{(j)}(0)$ is not divisible by p . If we take large enough p such that $p > n$ and $p > c_0$, then p also does not divide $nc_0 F(0)$ from equation(2).

Step IV. We now show that $\sum_{j=1}^r c_j \sum_{i=1}^n F(\sigma_i(\gamma_j))$ in (2) is an integer divisible by p . We do this by showing that each term $\sum_{i=1}^n F(\sigma_i(\gamma_j))$ is an integer divisible by p . Now,

$$\begin{aligned} \sum_{i=1}^n F(\sigma_i(\gamma_j)) &= \sum_{i=1}^n \sum_k f^{(k)}(\sigma_i(\gamma_j)) \\ &= \sum_k \sum_{i=1}^n f^{(k)}(\sigma_i(\gamma_j)). \end{aligned}$$

Since $g_j(x)^p$ divides $f(x)$ and each $\sigma_i(\gamma_j)$ is a root of $g_j(x)$, we see that

$$0 = f(\sigma_i(\gamma_j)) = f'(\sigma_i(\gamma_j)) = \cdots = f^{(p-1)}(\sigma_i(\gamma_j)).$$

For $t \geq p$, since $f^{(t)}(x) = p (b_1 \cdots b_r)^{pnr} h_t(x)$,

$$(8) \quad \sum_{i=1}^n f^{(t)}(\sigma_i(\gamma_j)) = p \cdot (b_1 \cdots b_r)^{pnr} \sum_{i=1}^n h_t(\sigma_i(\gamma_j)).$$

However, this above sum is invariant under the action of $\text{Gal}(K/\mathbb{Q})$, so it is a rational number. Moreover, $p(b_1 \cdots b_r)^{pnr} \sum_{i=1}^n h_t(x_i)$ is a symmetric polynomial in $x_1, \dots, x_n$ of degree at most $prn - 1$. By Symmetric polynomial theorem, the function is a polynomial in elementary symmetric functions of x_i . Putting $x_i = \sigma_i(\gamma_j)$ which are roots of the polynomial $g_j(x)$, with leading coefficient b_j , we get (3) is an integer. This shows that

$\sum_{j=1}^r c_j \sum_{i=1}^n F(\sigma_i(\gamma_j))$ in (2) is an integer divisible by p ; hence, the L.H.S in (2) is not congruent to 0 (mod p). This means that

$$\left| \sum_{j=1}^r \sum_{i=1}^n c_j e^{\sigma_i(\gamma_j)} \int_0^{\sigma_i(\gamma_j)} e^{-z} f(z) dz \right| \geq 1.$$

Step V. Let

$$\begin{aligned} m_1 &= \max_j \{|c_j|\}, \\ m_2 &= \max_{i,j} \left\{ \left| e^{\sigma_i(\gamma_j)} \right| \right\}, \\ m_3 &= \max_{i,j} \{|\sigma_i(\gamma_j)|\}, \end{aligned}$$

(we can take maximum of absolute values of finite set of elements in $\mathbb{C}$)
and

$$\begin{aligned} m_4 &= \max_{x \in [0,1]} \left\{ |e^{-z}| : z = x\sigma_i(\gamma_j) \right\}, \\ m_5 &= \max_{x \in [0,1]} \left\{ \prod_{j=1}^r |g_j(z)| : z = x\sigma_i(\gamma_j) \right\}, \end{aligned}$$

where m_4, m_5 exists as they are attained by continuous function on compact set. Now, we take the straight-line path from 0 to $\sigma_i(\gamma_j)$ as the curve for integration here. There we have the bound $|z^{p-1}| \leq |\sigma_i(\gamma_j)|^{p-1} \leq m_3^{p-1}$. This leads to the inequality

$$\begin{aligned} \left| \int_0^{\sigma_i(\gamma_j)} e^{-z} f(z) dz \right| &\leq m_3 m_4 \frac{(b_1 \cdots b_r)^{prn}}{(p-1)!} m_3^{p-1} m_5^p \\ &= m_4 \frac{(b_1 \cdots b_r)^{prn}}{(p-1)!} m_3^p m_5^p. \end{aligned}$$

Combining this with the previous inequality gives

$$\begin{aligned} 1 &\leq \left| \sum_{j=1}^r \sum_{i=1}^n c_j e^{\sigma_i(\gamma_j)} \int_0^{\sigma_i(\gamma_j)} e^{-z} f(z) dz \right| \\ &\leq r n m_1 m_2 m_4 \frac{(b_1 \cdots b_r)^{prn}}{(p-1)!} m_3^p m_5^p \\ &= r n m_1 m_2 m_4 \frac{((b_1 \cdots b_r)^{rn} m_3 m_5)^p}{(p-1)!}. \end{aligned}$$

Since for any $k \in \mathbb{R}$, we have $k^p/(p-1)! \rightarrow 0$ as $p \rightarrow \infty$, the last term in the inequality above can be made arbitrarily small by taking p large enough. This gives a contradiction. Therefore our assumption that the exponentials $e^{\alpha_1}, \dots, e^{\alpha_m}$ are linearly dependent over $\mathbf{Q}$ is false. This proves the theorem. $\square$

Corollary 9. *The numbers π and e are transcendental over $\mathbf{Q}$.*

Proof. Let e be algebraic over $\mathbf{Q}$. Then there are rationals r_i such that $\sum_{i=0}^n r_i e^i = 0$. Then the numbers $e^0, e^1, \dots, e^{n-1}$ are linearly dependent over $\mathbf{Q}$. By choosing $m = n + 1$ and $\alpha_i = i - 1$, this dependence is false by the above theorem. Thus, e is transcendental over $\mathbf{Q}$. For π , we note that if π is algebraic over $\mathbf{Q}$, then so is πi ; hence, $e^0, e^{\pi i}$ are linearly independent over $\mathbf{Q}$, which is false since $e^{\pi i} = -1$. Thus, π is transcendental over $\mathbf{Q}$. $\square$